Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 3A nr. 40.8

$$f(x) = e^{\frac{\sqrt[3]{\sin x}}{x}}$$

$$f'(x) = e^{\frac{\sqrt[3]{\sin x}}{x}} \cdot \left(\frac{\sqrt[3]{\sin x}}{x}\right)'$$

$$\left(\frac{\sqrt[3]{\sin x}}{x}\right)' = \frac{x \cdot (\sqrt[3]{\sin x})' - \sqrt[3]{\sin x} \cdot (x)'}{x^2}$$

$$(\sqrt[3]{\sin x})' = \left((\sin x)^{\frac{1}{3}}\right)' = \frac{1}{3}(\sin x)^{-\frac{2}{3}} \cdot \cos x = \frac{\cos x}{3\sqrt[3]{\sin^2 x}}$$

$$\left(\frac{\sqrt[3]{\sin x}}{x}\right)' = \frac{x \cdot \frac{\cos x}{3\sqrt[3]{\sin^2 x}} - \sqrt[3]{\sin x}}{x^2}$$

$$\left(\frac{\sqrt[3]{\sin x}}{x}\right)' = \frac{\frac{x \cos x - 3\sqrt[3]{\sin x} \cdot \sqrt[3]{\sin^2 x}}{3\sqrt[3]{\sin^2 x}}}{x^2}$$

$$\left(\frac{\sqrt[3]{\sin x}}{x}\right)' = \frac{x \cos x - 3 \sin x}{3x^2 \sqrt[3]{\sin^2 x}}$$

$$f'(x) = e^{\frac{\sqrt[3]{\sin x}}{x}} \cdot \frac{x \cos x - 3 \sin x}{3x^2 \sqrt[3]{\sin^2 x}}$$

References